

Design, Simulation, and Hardware Implementation of a Quadrature Down Converter

Comprehensive Project Report & Laboratory Verification Presentation

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Presentation Outline

System Specifications & Design Constraints

Project Performance Requirements & Directives

- **Technology Node & Models:** TSMC 180nm CMOS technology for NMOS switches and UA741 operational amplifiers.
- **MOSFET Parasitic Modeling:** Geometric layout parameter optimization: W, L, A_S, A_D, P_S, P_D .
- **Amplifier Architecture Choice:** Implemented a discrete **CE BJT Amplifier** stage with emitter degeneration for low-distortion output voltage gain.
- **Oscillator Target:** Quadrature v_{OSC_I} & v_{OSC_Q} at 170 kHz with 90° phase shift and 1 V_{pp} swing.
- **Filter & Output Spec:** Cascaded LPF cutoff $f_c \approx 4.65$ kHz, final amplified IF output ≥ 400 mV_{pp} at 5 kHz.

Utility & Need for Quadrature (I-Q) Operation

- **Need in Wireless Receivers (Bluetooth, Wi-Fi, WLAN, 5G):**

- Modern digital modulation (QAM, PSK) transmits complex vector signals:

$$s(t) = I(t) \cos(2\pi f_c t) - Q(t) \sin(2\pi f_c t)$$

- Doubles spectral bandwidth efficiency compared to real-only modulation.

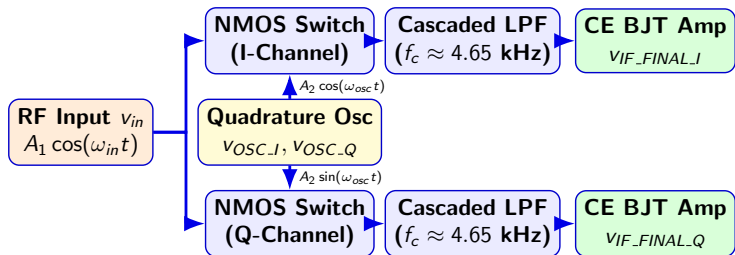
- **Image Rejection Without Off-Chip RF Filters:**

- Direct-conversion / low-IF receivers reject image frequencies via 90° phase cancellation, enabling 100% CMOS single-chip integration.

- **Solving the Negative Frequency Overlap Problem:**

- Real-only mixing with $\cos(\omega_c t)$ causes upper and lower sidebands to fold onto each other. Quadrature mixing uniquely separates magnitude and phase.

QDC System Architecture & Mathematical Derivations



Quadrature Downconversion Mathematical Equations

$$v_{IF_I} = v_{in} \times v_{OSC_I} = \frac{A_1 A_2}{2} [\cos((\omega_{in} - \omega_{osc})t) + \cos((\omega_{in} + \omega_{osc})t)]$$

$$\xrightarrow{\text{After LPF}} v_{IF_FINAL_I} = \frac{A_1 A_2}{2} \cos(\omega_{IF} t)$$

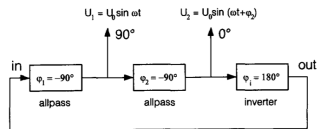
$$v_{IF_Q} = v_{in} \times v_{OSC_Q} = \frac{A_1 A_2}{2} [\sin((\omega_{in} + \omega_{osc})t) - \sin((\omega_{in} - \omega_{osc})t)]$$

$$\xrightarrow{\text{After LPF}} v_{IF_FINAL_Q} = \frac{A_1 A_2}{2} \sin(\omega_{IF} t)$$

Quadrature Oscillator Topology Selection

3-Op-Amp Allpass Quadrature Topology:

- Consists of two active **allpass filter stages** and one **inverting amplifier** with soft nonlinear feedback.
- Why 3-Op-Amp Allpass over 2-Op-Amp Integrator?**
 - Wide Bandwidth:* Allpass filters maintain unity gain ($G = 1$) across high frequencies up to tens of MHz.
 - Precise Quadrature:* Sampling between identical stages guarantees an exact 90° phase shift.
 - Amplitude Stability:* Uniform loop gain prevents envelope variation during frequency adjustments.



Allpass Quadrature Oscillator Block Diagram.

Oscillator Derivations & Component Calculations

- **Allpass Phase Shift Transfer Function:**

$$H(s) = \frac{sRC - 1}{sRC + 1} \implies \phi(f) = -2 \arctan(2\pi fRC)$$

- **Quadrature Oscillation Condition ($\phi = -90^\circ$ per stage):**

$$2\pi f_{\text{osc}} RC = 1 \implies f_{\text{osc}} = \frac{1}{2\pi RC}$$

- **Component Calculations for Target Frequencies ($C = 27 \text{ pF}$):**

- For $f_{\text{osc}} = 150 \text{ kHz}$: $R = \frac{1}{2\pi(150 \text{ k})(27 \text{ p})} \approx 39.3 \text{ k}\Omega$
- For $f_{\text{osc}} = 200 \text{ kHz}$: $R = \frac{1}{2\pi(200 \text{ k})(27 \text{ p})} \approx 29.5 \text{ k}\Omega$

- **Supply Voltages (V_{DD}, V_{SS}):** $\pm 15 \text{ V}$ supplies provide wide linear operating range for standard UA741 op-amps to yield the requested 1 V_{pp} ($500 \text{ mV}_{\text{peak}}$) swing.

Diode Feedback & Asymmetry Justification

Nonlinear Diode Network

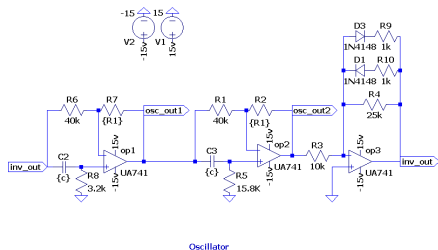
- Diodes (D1, D3) and resistors (R9, R10) in inverter (op3) feedback path.
- **Startup:** Initial gain > 1 ($R_4/R_3 > 1$) ensures rapid startup from noise.
- **Soft Clamping:** At $1 V_{pp}$, diodes conduct, reducing effective gain to lock $A\beta = 1$.

Justification of Asymmetric Tuning

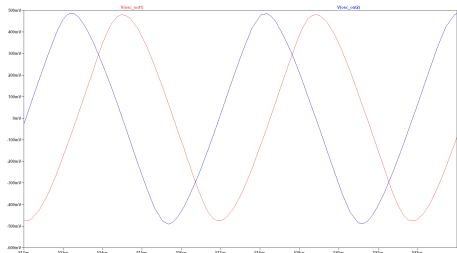
- Symmetric SPICE values led to waveform distortion in physical breadboard realization.
- UA741 high-frequency internal poles introduce parasitic phase lag at > 150 kHz.
- Asymmetric tuning ($R_8 = 432 \Omega$, $R_5 = 15.2 \text{ k}\Omega$) compensates for parasitics and stage loading.

LTSpice Simulation – Schematic & Transient Waveforms

```
.include UA741.301
.param c 27p
.param R1=50k
.tran 0 2m 0 1u
```



(a) Complete UA741 Quadrature Oscillator Schematic.



(b) Transient Simulation: Period $T = 5.424 \mu s$, phase shift $= -268.49^\circ (+91.51^\circ)$.

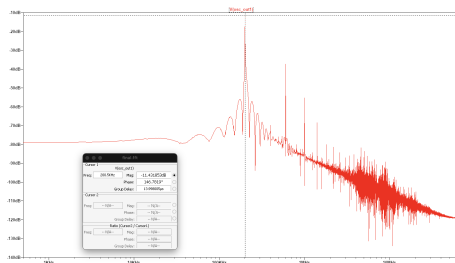
Verification: Output swing stabilizes precisely at $1 V_{pp}$ ($500 mV_{peak}$) with clean 90° phase shift.

Phase Measurement & FFT Spectral Analysis

```
SPICE Output Log: C:\users\bhargav_vemparala\Documents\LTspice\project\final.log
C:\users\bhargav_vemparala\Documents\LTspice\project\cd4007.lib

t1: V(osc_out1)=0 AT 0.00499588611483
t2: V(osc_out2)=0 AT 0.00499993145422
tp1: V(osc_out1)=0 AT 0.000551806464335
tp2: V(osc_out1)=0 AT 0.000557230642398
period: (tp2-tp1)=5.42417806316e-06
phasediff: -(t2-t1)*360/period=-268.487163141
```

SPICE Error Log phase calculation verifying 91.51° phase shift
between I and Q.

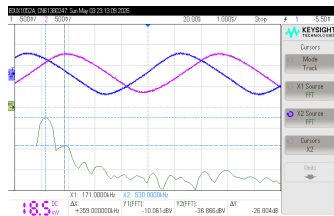
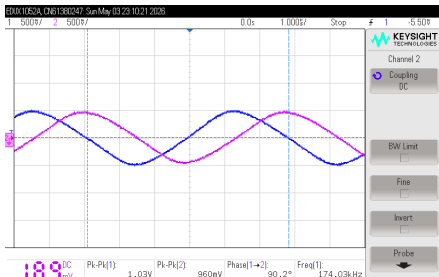


FFT Spectrum of v_{OSC_I} : Strong peak at 200.5 kHz,
magnitude -11.43 dB, phase 146.78° .

Hardware Realization (741 ICs) & Parameter Tweaks

Table: Resistor Comparison: SPICE vs. Lab Potentiometer Method

Component	SPICE Value (Ω)	Real Life Value (Ω)
R_1/R_7	40 k/50 k	50 k/47 k
R_2/R_6	50 k/40 k	47 k/39.2 k
R_3/R_4	10 k/25 k	10 k/17.6 k
R_5/R_8	15.8 k/3.2 k	15.2 k/432
R_9/R_{10}	1 k/1 k	2.2 k/2.2 k



confirming hardware spectral purity.

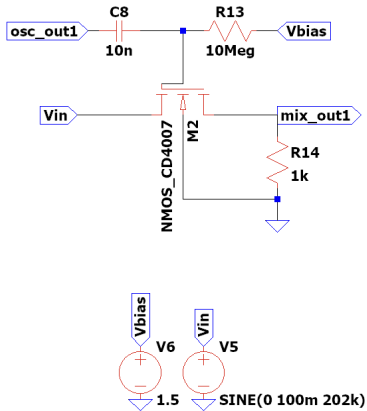
Switch (Mixer) Topology & MOSFET Optimization

- **Active NMOS Switch Topology:** Passive NMOS switch operating in deep triode region.
 - RF Input ($v_{in} = 100$ mV) connected to Drain; LO (v_{OSC}) AC-coupled to Gate.
 - Intermediate Frequency output (v_{IF}) taken at Drain across $R_L = 1$ k Ω .
- **MOSFET Parameter Optimization (180 nm CMOS Node):**
 - Channel Length: $L = 0.18$ μm (minimum feature size).
 - Width $W = 20$ μm for target $R_{on} \approx 50$ Ω ($V_{GS} - V_{th} = 0.6$ V):

$$\frac{W}{L} = \frac{1}{R_{on} \cdot \mu_n C_{ox} (V_{GS} - V_{th})} = \frac{1}{50 \cdot 300 \mu \cdot 0.6} \approx 111.1$$
$$\implies W = 111.1 \times 0.18 \mu\text{m} \approx \mathbf{20 \mu\text{m}}$$

- **LTspice Geometric Parasitics ($Y = 0.18$ μm):**
 - Source/Drain Area: $A_S = A_D = W \times Y = 3.6$ p (3.6p)
 - Source/Drain Perimeter: $P_S = P_D = W + 2Y = 20.36$ μm (20.36u)

Mixer LTSpice Schematic & Component Values



Mixer

LTSpice NMOS Switch Mixer Schematic.

Table: Mixer Component Summary

Component	Value	Rationale
M_1	NMOS (20u/0.18u)	Core switching element.
R_{BIAS}	10 M Ω	Prevents AC loading on LO gate node.
C_C	10 nF	AC couples LO signal; blocks DC.
R_L	1 k Ω	Standard load for gain/bandwidth balance.
V_{BIAS}	0.6 V – 0.8 V	Sets gate near V_{th} for sharp switching.

- **Square-Wave Switching Expansion:**

$$s(t) = \frac{1}{2} + \frac{2}{\pi} \sum_{n=1,3,5\dots}^{\infty} \frac{\sin(n\omega_{\text{osc}}t)}{n}$$

- **Product-to-Sum Multiplication:**

$$\begin{aligned} v_{\text{out}}(t) &\approx A_{\text{in}} \cos(\omega_{\text{in}}t) \cdot \frac{2}{\pi} \sin(\omega_{\text{osc}}t) \\ &= \frac{A_{\text{in}}}{\pi} [\sin((\omega_{\text{osc}} - \omega_{\text{in}})t) + \sin((\omega_{\text{osc}} + \omega_{\text{in}})t)] \end{aligned}$$

- **Extracted Frequencies ($f_{\text{osc}} = 170 \text{ kHz}$, $f_{\text{in}} = 165 \text{ kHz}$):**

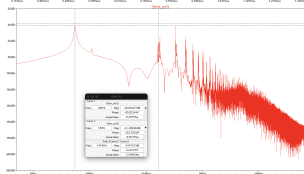
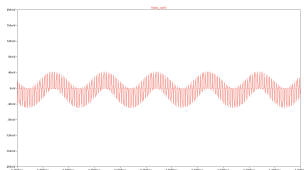
- **Downconverted IF Component:**

$$f_{\text{IF}} = |f_{\text{osc}} - f_{\text{in}}| = |170 \text{ k} - 165 \text{ k}| = 5 \text{ kHz}$$

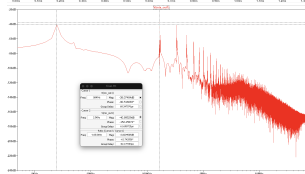
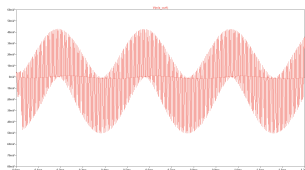
- **Sum Component (Filtered by LPF):**

$$f_{\text{sum}} = 170 \text{ k} + 165 \text{ k} = 335 \text{ kHz}$$

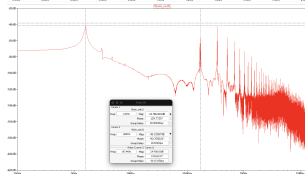
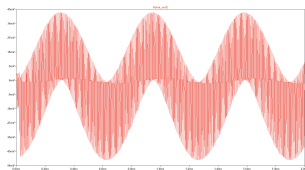
Frequency Sweep Analysis ($f_{IN} < f_{OSC}$: 165–169 kHz)



165 kHz Input: $f_F = 5$ kHz

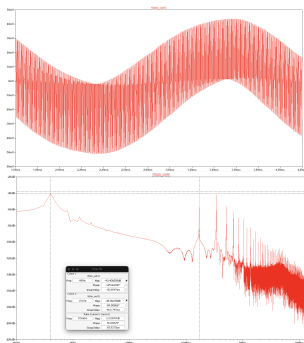


168 kHz Input: $f_F = 2$ kHz

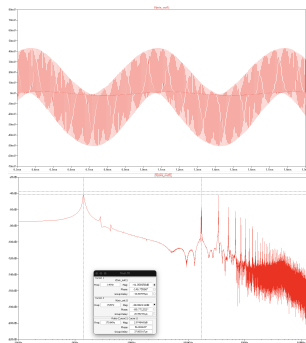


169 kHz Input: $f_F = 1$ kHz

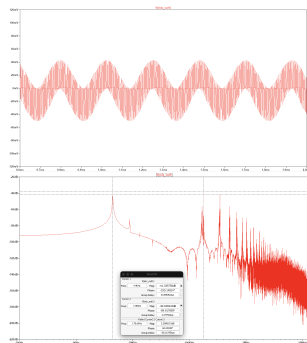
Frequency Sweep Analysis ($f_{IN} > f_{OSC}$: 171–175 kHz)



171 kHz Input: $f_F = 1$ kHz

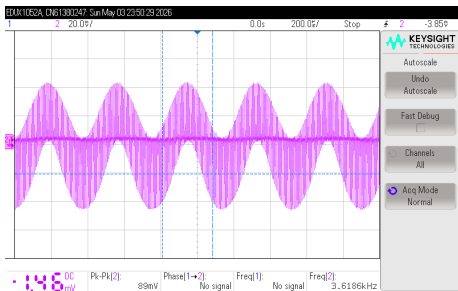


172 kHz Input: $f_F = 2$ kHz

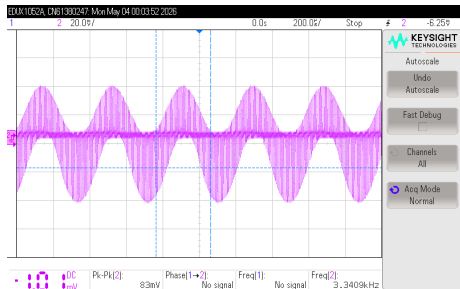


175 kHz Input: $f_F = 5$ kHz

Mixer Lab Realization & DSO Waveform Captures



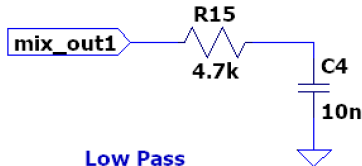
DSO Measurement 1: Hardware NMOS mixer output envelope.



DSO Measurement 2: High-frequency carrier with low-frequency IF envelope.

Result: Laboratory NMOS hardware confirms successful frequency downconversion matching SPICE predictions.

Low-Pass RC Filter Design & Cutoff Calculations



First-Order RC Low-Pass Filter Schematic.

- **Cutoff Frequency Formula:**

$$f_c = \frac{1}{2\pi RC}$$

- **Lab Practical Implementation:**

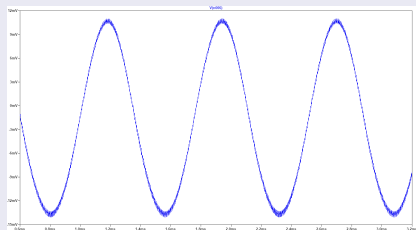
- $R = 4.7 \text{ k}\Omega$, $C = 10 \text{ nF}$
- $f_c = \frac{1}{2\pi(4700)(10 \text{ n})} \approx 3.39 \text{ kHz}$

- **Theoretical 3 kHz Target Resistor:**

$$R_{\text{ideal}} = \frac{1}{2\pi(3000)(10 \text{ nF})} \approx 5.3 \text{ k}\Omega$$

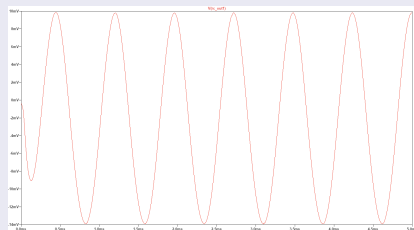
LPF Optimization – 1-Stage vs. 2-Stage Cascaded LPF

Iteration 1: Single-Stage LPF



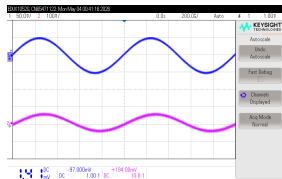
- Roll-off: -20 dB/decade.
- **Drawback:** High-frequency switching noise and 335 kHz sum harmonic bleed through.

Iteration 2: Two-Stage Cascaded LPF

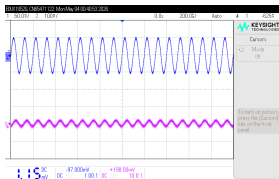


- Components: $2 \times (2.2 \text{ k}\Omega, 10 \text{ nF})$.
- Roll-off: -40 dB/decade.
- $f_{c,\text{cascaded}} \approx 7.23 \text{ k} \times \sqrt{\sqrt{2} - 1} \approx 4.65 \text{ kHz}$.

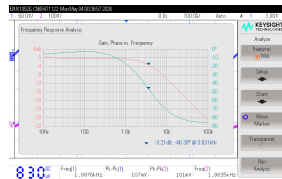
LPF Hardware Code Plot & Transient Response



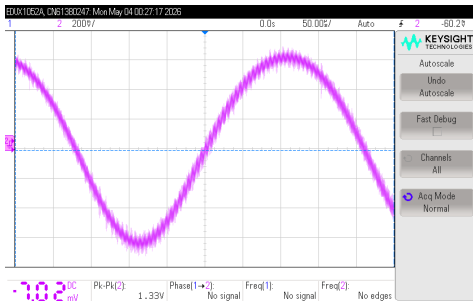
1 kHz Passband Response



10 kHz Attenuated Response



Hardware Code Plot (AC Response)



DSO Capture: Integrated Mixer + Cascaded LPF Output (5 kHz IF Tone).

CE BJT Amplifier Design (Discrete Gain Stage)

- **Amplifier Topology Selection:** Implemented a discrete **Common Emitter (CE) BJT Amplifier** for output signal boosting.
- **Problem Statement:** Filter attenuation reduces 5 kHz IF to ~ 8 mV. Target specification requires ≥ 400 mV_{pp}.
- **Emitter Degeneration ($R_{E1} = 15\ \Omega$) for Low THD:**

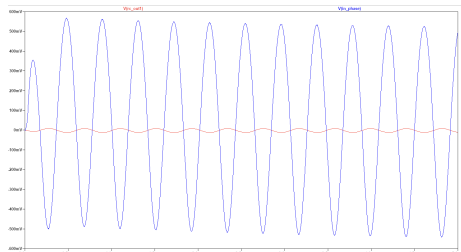
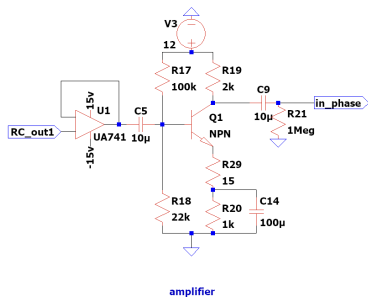
$$I_E \approx 1.46\text{ mA} \implies r_e = \frac{V_T}{I_E} \approx \frac{26\text{ mV}}{1.46\text{ mA}} \approx 17.8\ \Omega$$

$$A_v \approx \frac{R_C}{R_{E1} + r_e} = \frac{2000}{15 + 17.8} \approx \mathbf{61}$$

$$V_{\text{out}} = V_{\text{in}} \times A_v \approx 8\text{ mV} \times 61 \approx \mathbf{488\text{ mV}_{\text{pp}}}$$

- **Result:** 488 mV_{pp} swing comfortably satisfies the ≥ 400 mV_{pp} specification.

Amplifier Lab Setup & Output Swing

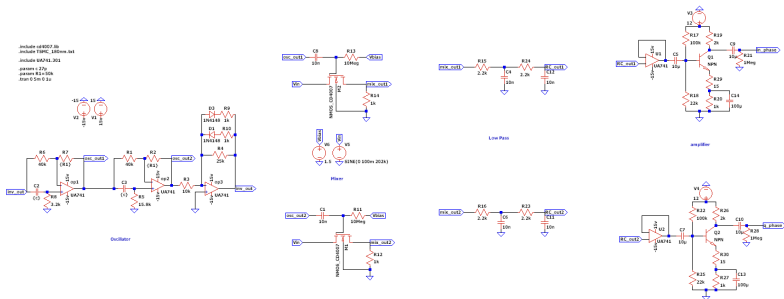


(b) Integrated DSO Output: Mixer + LPF + Amplifier (5 kHz, > 450 mV_{pp}).

(a)

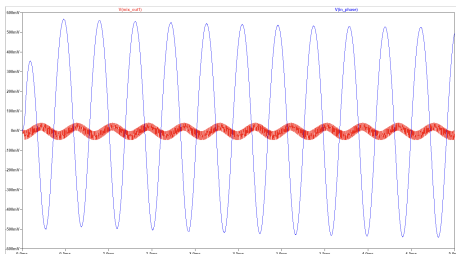
Hardware Breadboard Setup of CE BJT Amplifier.

Complete Integrated QDC Prototype Schematic

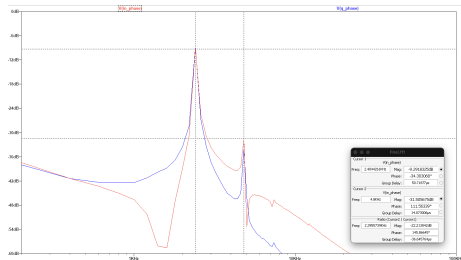


- Complete QDC architecture integrating Quadrature Oscillator, dual NMOS Mixers, dual Cascaded LPFs, and dual CE BJT Amplifiers.

System LTSpice Simulation Results



(a) Transient Simulation of Final $v_{IF_FINAL_I}$ and $v_{IF_FINAL_Q}$ Signals (5 kHz, 90°).

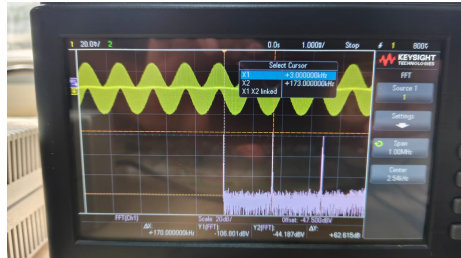


(b) FFT Spectrum of Final I and Q Outputs showing pure 5 kHz peaks.

Hardware DSO Measurements – Single & Envelope

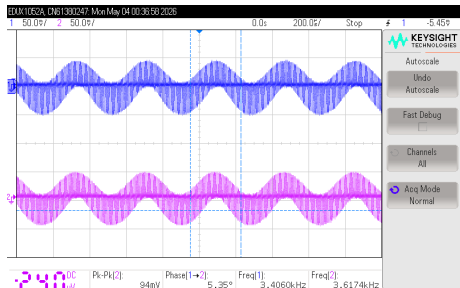
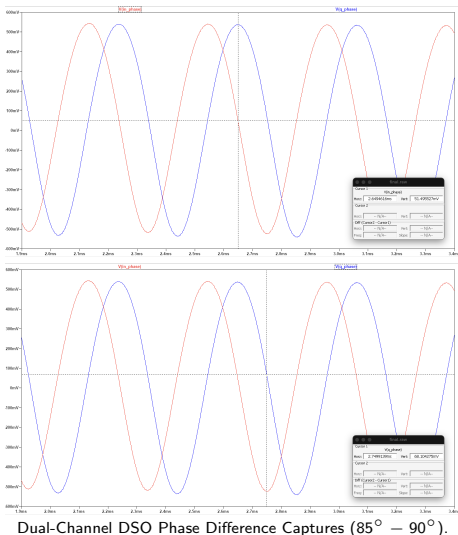


(a) DSO Measurement: Single IF Channel Output (5 kHz).



(b) DSO Measurement: IF Output overlaid with 170 kHz RF Input Envelope.

Hardware Quadrature Phase Shift Verification



Final DSO Display: Concurrent In-Phase (I) and Quadrature (Q) IF Outputs.

Performance Summary & Quantitative Comparison

Table: System Performance Summary: Simulated vs. Measured

Parameters	Simulated	Measured
Oscillator Frequency (f_{osc})	200 kHz	174 kHz
Oscillator Amplitude (I-phase)	500 mV _{peak}	430 mV _{peak}
Oscillator Amplitude (Q-phase)	500 mV _{peak}	515 mV _{peak}
Input frequency (f_{in})	202 kHz	171 kHz
Intermediate Frequency (f_{IF})	2 kHz	3 kHz – 5 kHz
Supply Voltages	± 15 V	± 12 V
Mixer Gate Bias (V_{BIAS})	1.2 V	1.2 V
Mixer Coupling Cap (C_C)	10 nF	10 μ F
Phase Difference (I vs Q)	93°	85°
Final IF Amplitude (V_{IF_FINAL})	1.3 V _{pp}	800 mV _{pp} (≥ 400 mV _{spec})
LPF Components (R, C)	4.7 k Ω , 10 nF	2 $\times$ (2.2 k Ω , 10 nF)

Conclusion & Group Member Contributions

Project Accomplishments

- Full implementation and laboratory verification of an analog Quadrature Down Converter operating at 170 kHz with a 5 kHz IF output.
- Successfully verified TSMC 180nm CMOS active mixer, 3-op-amp allpass quadrature oscillator, 2-stage cascaded LPF, and discrete CE BJT amplifier stage.

Group Member Contributions

- **Katta Sri Kaushik (2025102071):** Quadrature Oscillator design, allpass phase calculations, diode soft limiting network tuning, and LTSpice transient/FFT simulations.
- **Lakshmi Sai Bhargav V (2025102061):** Active NMOS switch mixer design, MOSFET sizing (W/L , A_S , P_S), frequency sweep simulations, and integrated system layout.
- **Kushwanth Lanka (2025102072):** Two-stage cascaded RC LPF optimization, CE BJT amplifier design with emitter degeneration for low THD, and physical DSO hardware testing.

Thank You!

Questions & Technical Discussion